MATH 220 — Differential Equations

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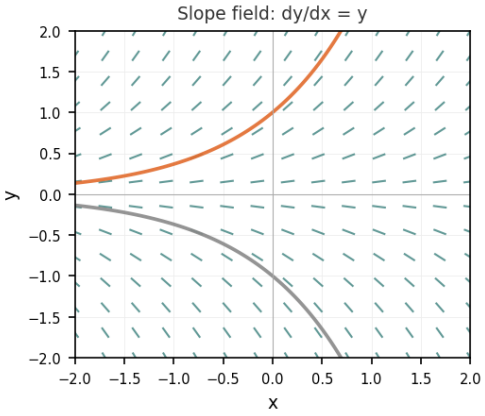
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Learning Objectives

- Read and interpret slope fields; sketch solution curves through given points
- Solve separable differential equations with initial conditions
- Model exponential growth and decay by setting up and solving dP/dt = kP
- Apply Newton's Law of Cooling and the logistic growth model

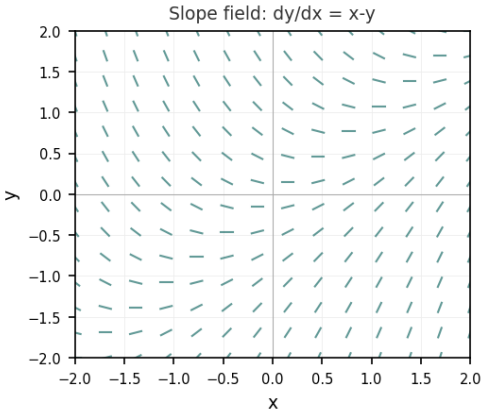
For separable DEs: separate variables completely before integrating. Always apply the initial condition AFTER finding the general solution.

1. The slope field below is for dy/dx = y. Sketch the solution curve through (0, 1). What is the general solution?



Answer: _____

2. Analyze the slope field below (dy/dx = x − y). Describe where slopes are zero, positive, and negative.



Answer: _____

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3. Solve the separable differential equation with initial condition.

$$\frac{dy}{dx} = 3x^2y, \quad y(0) = 2$$

Answer: _____

4. Solve the separable differential equation.

$$\frac{dy}{dx} = \frac{x^2}{y^2}, \quad y(0) = 1$$

Answer: _____

5. Solve the separable DE.

$$\frac{dy}{dx} = e^{x-y} = \frac{e^x}{e^y}$$

Answer: _____

6. A bacterial population grows at a rate proportional to its size. Write and solve the differential equation. The population doubles in 3 hours, starting at P0 = 500.

$$\frac{dP}{dt} = kP, \quad P(0) = 500, \quad P(3) = 1000$$

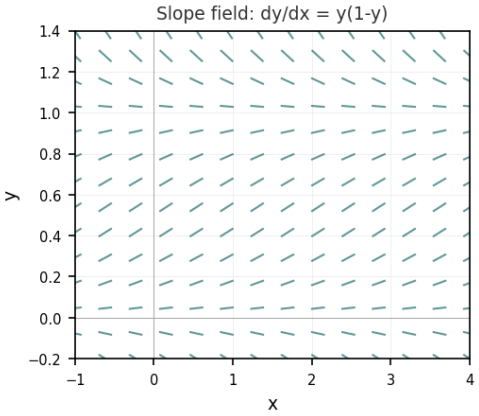
Answer: _____

7. Newton's Law of Cooling: $dT/dt = -k(T - T_s)$. A cup of coffee at 90°C cools in a 20°C room. After 5 minutes, it is 70°C. Find T(t) and when it reaches 40°C.

$$\frac{dT}{dt} = -k(T - 20), \quad T(0) = 90, \quad T(5) = 70$$

Answer: _____

8. The logistic growth model is $dP/dt = rP(1 - P/K)$. With r = 1 and K = 1, the slope field is shown. Identify equilibrium solutions and describe population behavior.



Answer: _____

9. Solve the first-order linear differential equation.

$$\frac{dy}{dx} + 2y = 4x, \quad y(0) = 1$$

Answer: _____

10. Verify that y = x² + Ce^(–x) is a solution to dy/dx + y = 2x + x².

$$\frac{dy}{dx} + y = 2x + x^2$$

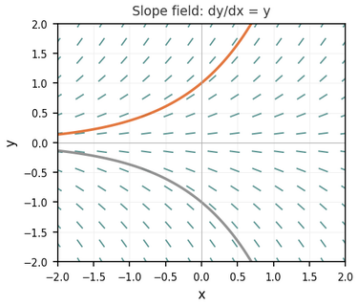
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Problems 1–2 are slope field interpretation — build visual intuition before algebra. Problem 6 (bacterial growth): emphasize writing the DE first, then solving. Problem 8 (logistic): equilibrium analysis is more important than the explicit solution.

Solutions

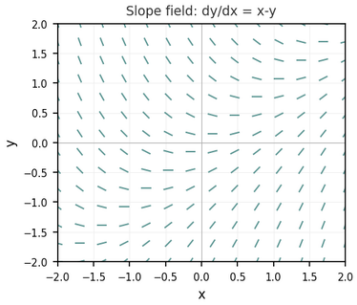
1. The slope field below is for dy/dx = y. Sketch the solution curve through (0, 1). What is the general solution?



→ *dy/dx = y is a separable equation: dy/y = dx.*
→ *Integrate both sides: ln|y| = x + C_1 → y = Ce^x.*
→ *Initial condition y(0)=1: 1=Ce^0=C → C=1.*
→ *Solution: y = e^x. The orange curve in the slope field.*

Answer: $y = Ce^x, \quad \text{through } (0,1): \; y = e^x$

2. Analyze the slope field below (dy/dx = x – y). Describe where slopes are zero, positive, and negative.



→ *Slopes are zero (dy/dx=0) when x=y — along the line y=x.*
→ *Slopes are positive when x > y (above the line y=x: dy/dx>0).*
→ *Slopes are negative when x < y.*
→ *This is a first-order linear DE. Integrating factor: e^x.*
→ *General solution: y = x – 1 + Ce^{-x}.*

Answer: Nullcline: $x = y$ (zero slope along $y = x$)

3. Solve the separable differential equation with initial condition.

$$\frac{dy}{dx} = 3x^2y, \quad y(0) = 2$$

→ *Separate: dy/y = 3x² dx.*
→ *Integrate: ln|y| = x³ + C.*
→ *y = Ae^{x³}. Apply y(0)=2: A=2.*
→ *Solution: y = 2e^{x³}.*

Answer: $y = 2e^{x^3}$

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4. Solve the separable differential equation.

$$\frac{dy}{dx} = \frac{x^2}{y^2}, \quad y(0) = 1$$

→ *Separate: y² dy = x² dx.*
→ *Integrate: y³/3 = x³/3 + C.*
→ *y³ = x³ + K. Apply y(0)=1: 1=K.*
→ *y³ = x³+1 → y = (x³+1)^{1/3}.*

Answer: $y = (x^3 + 1)^{1/3}$

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5. Solve the separable DE.

$$\frac{dy}{dx} = e^{x-y} = \frac{e^x}{e^y}$$

→ *Separate: e^y dy = e^x dx.*
→ *Integrate: e^y = e^x + C.*
→ *y = ln(e^x + C).*

Answer: $e^y = e^x + C \Rightarrow y = \ln(e^x + C)$

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6. A bacterial population grows at a rate proportional to its size. Write and solve the differential equation. The population doubles in 3 hours, starting at P0 = 500.

$$\frac{dP}{dt} = kP, \quad P(0) = 500, \quad P(3) = 1000$$

→ *Separable: dP/P = k dt -> ln P = kt + C -> P = P0·e^{kt}.*
→ *P(0)=500: P₀=500.*
→ *P(3)=1000: 500e^{3k}=1000 → e^{3k}=2 → k=ln2/3.*
→ *P(t) = 500·e^{(ln2/3)t} = 500·2^{t/3}.*
→ *Find P(9): 500·2³ = 4000 bacteria.*

Answer: $P(t) = 500\,e^{(\ln 2/3)\,t}$

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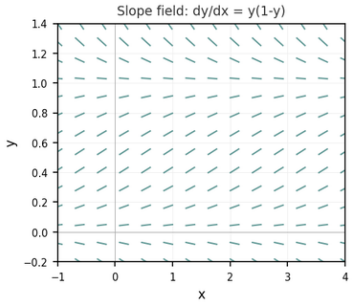
7. Newton's Law of Cooling: dT/dt = –k(T – T_s). A cup of coffee at 90°C cools in a 20°C room. After 5 minutes, it is 70°C. Find T(t) and when it reaches 40°C.

$$\frac{dT}{dt} = -k(T - 20), \quad T(0) = 90, \quad T(5) = 70$$

→ *Let u=T–20. Then du/dt=–ku → u=u₀e^{–kt}.*
→ *T(t) = 20+70e^{–kt}. At t=5: 70=70e^{–5k}... wait.*
→ *T(5)=70 → 70=20+70e^{–5k} → 50=70e^{–5k} → e^{–5k}=5/7.*
→ *k=–(1/5)ln(5/7)=(1/5)ln(7/5).*
→ *T=40: 40=20+70e^{–kt} → e^{–kt}=2/7 → t=ln(7/2)/k.*

Answer: $T(t) = 20 + 70e^{-kt}, \quad k = \frac{1}{5}\ln\frac{7}{5}$

8. The logistic growth model is dP/dt = rP(1 – P/K). With r = 1 and K = 1, the slope field is shown. Identify equilibrium solutions and describe population behavior.



→ *Set dP/dt=0: P(1–P)=0 → P=0 or P=1.*
→ *P=0 is an unstable equilibrium (populations grow away from 0).*
→ *P=1 (carrying capacity K=1) is a stable equilibrium.*
→ *Populations below K increase toward K; populations above K decrease toward K.*
→ *S-shaped (sigmoidal) solution curves are visible in the slope field.*

Answer: $P = 0$ (unstable), $P = 1$ (stable carrying capacity)

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9. Solve the first-order linear differential equation.

$$\frac{dy}{dx} + 2y = 4x, \quad y(0) = 1$$

→ *Standard form: dy/dx + P(x)y = Q(x). Here P=2, Q=4x.*
→ *Integrating factor: μ = $e^{\int 2dx} = e^{2x}$.*
→ *Multiply: d/dx[e^{2x}y] = 4xe^{2x}.*
→ *Integrate RHS by parts: ∫4xe^{2x}dx = 2xe^{2x}–e^{2x}+C.*
→ *e^{2x}y = 2xe^{2x}–e^{2x}+C → y = 2x–1+Ce^{-2x}.*
→ *y(0)=1: 1=0–1+C → C=2. y=2x–1+2e^{-2x}.*

Answer: $y = 2x - 1 + 2e^{-2x}$

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10. Verify that y = x² + Ce^(–x) is a solution to dy/dx + y = 2x + x².

$$\frac{dy}{dx} + y = 2x + x^2$$

→ *Compute dy/dx: dy/dx = 2x – Ce^{-x}.*
→ *LHS = dy/dx + y = (2x–Ce^{-x}) + (x^2+Ce^{-x}).*
→ *= 2x – Ce^{-x} + x^2 + Ce^{-x} = 2x + x^2.*
→ *LHS = RHS = 2x + x^2. ✓ y = x^2+Ce^{-x} is the general solution.*

Answer: Verified: $\text{LHS} = 2x + x^2 = \text{RHS}$

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